Name: _____

Class: _____Date:_____

LESSON 8. DIVISIBILITY RELATION AND ITS PROPERTIES

2.1. Replace each ? with a suitable symbol: ⋮ (is divisible by) or $\not\vdots$ (is not divisible by).

72 [?] 8; 54 [?] 7; 1 875 [?] 25; 2 030 [?] 14; 2 027 [?] 3.

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2.2. Find all the divisors of 54.

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2.3. Find the multiples of 7 that are greater than 50 and less than 100.

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2.4. Which statements below are true? Why?

a) $2\,024 \cdot 13 + 12$ is divisible by 13;

b) $85 \cdot 24 + 6$ is divisible by 6;

c) $2\,030 \cdot 40 + 6 \cdot 15$ is divisible by 10.

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2.5. Without calculating, decide which sum below is divisible by 5:

a) 95 + 2 030 + 25; b) 1 940 + 200 + 2 023.

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2.6. Using the divisibility property of a sum, find x in the set {25; 31; 60; 48} such that x + 35 is divisible by 5.

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2.7. Using the divisibility property of a difference, find x in the set {21; 34; 57; 80} such that x − 9 is divisible by 3.

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2.8. Using the divisibility property of a sum, find x in the set {24; 30; 44; 63} such that x + 36 is **not** divisible by 4.

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2.9. a) Why is the sum $3^2 + 3^3 + 3^4 + 3^5$ divisible by 4?

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b) Why is the sum $6^{30} + 6^{31} + 6^{32} + 6^{33}$ divisible by 7?

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2.10. When the natural number a is divided by 15, the remainder is 10. Is a divisible by 5? Is a divisible by 3?

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2.11. To open a safe, Mai needs an 8-digit code made by joining 4 two-digit numbers from the table below. These are the numbers that are divisible by 4 or divisible by 5, written in order from smallest to largest. Help Mai open the safe!

17	36	41	58	29	45
53	62	71	86	94	39
23	66	51	70	98	92

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